Professional Self-Assessment — Andre Miranda

Throughout my Computer Science program and the development of my CS-499 ePortfolio, I have grown into a more capable, confident, and technically versatile professional. As a Product Owner who also contributes directly as a developer, engineer, and tester, this program has strengthened my ability to bridge the gap between product vision and technical execution. The coursework, enhancements, and capstone experience have allowed me to demonstrate not only my technical competence, but also the professional values that define how I work. These include but are not limited to security, maintainability, clean architecture, user-centered design, reliability, teamwork, and integrity. These values guide my decisions as a product leader and shape the way I build and support modern software systems.

The Computer Science program has helped me refine my professional goals and expand my capabilities beyond product strategy. I now have the technical depth to contribute meaningfully to engineering discussions, evaluate architectural tradeoffs, design secure and maintainable systems, and collaborate more effectively with developers and stakeholders. This combination of product ownership and hands-on engineering skill is central to my long-term goal of becoming a Director or VP in the technology space. I want to lead teams not only with vision, but with a grounded understanding of how high-quality software is built, tested, secured, and maintained.

Collaboration has been a consistent theme throughout my professional life and career. These experiences strengthened my ability to work in team environments, communicate clearly, and translate technical concepts into actionable requirements, skills that are essential for a Product Owner. Communicating with stakeholders has also been a core part of my development. Whether writing enhancement plans, producing narratives, or presenting code reviews, I learned how to articulate technical decisions in a way that is accessible, structured, and aligned with business goals.

The program also deepened my understanding of algorithms and data structures. I learned how to design efficient solutions, evaluate tradeoffs, and apply algorithmic thinking to real problems. This knowledge became especially important during my capstone enhancements, where I implemented a weighted Rescue Suitability Scoring Engine using caching, normalization, aggregation, and ranking logic. These experiences strengthened my ability to reason about performance, complexity, and data-driven decision making. These skills directly support my work as both a product owner and technical contributor.

In software engineering and database technologies, I gained hands-on experience designing modular systems, implementing secure CRUD operations, optimizing database

queries, and applying defensive programming techniques. I learned how to build maintainable architectures, use environment-based configuration, apply structured logging, and design systems that are resilient and scalable. Security was emphasized throughout the program, and I consistently applied secure coding practices such as input validation, operator whitelisting, authentication enforcement, and safe error handling. These practices align with my professional values and reflect the mindset I bring to every project.

The artifacts in my ePortfolio demonstrate the full range of these skills. Together, they form a cohesive narrative of growth across software engineering, algorithms, databases, security, testing, and UI/UX design. The enhanced CRUD module showcases secure programming, modular architecture, and professional engineering practices. The Rescue Suitability Scoring Engine highlights algorithmic thinking, data processing, and decision support logic. The enhanced dashboard demonstrates full-stack development, database integration, analytics, visualization, and user-centered design. The testing suite reflects my commitment to reliability, correctness, and defensive programming. Each artifact builds on the others, illustrating how I approach problems holistically from backend logic to frontend experience, from database design to product usability.

Completing the capstone and assembling this ePortfolio has helped me articulate my strengths, refine my professional identity, and prepare to advance further in the technology field. I now have a portfolio that demonstrates not only what I can build, but how I think, how I collaborate, and how I lead. It reflects my readiness to contribute as a Product Owner who understands engineering deeply, and as a future technology leader committed to building secure, maintainable, user-centered, and reliable systems. This program has shaped my goals, strengthened my values, and prepared me to be more employable, more effective, and more impactful in the computer science profession.